Passing Is Not Shipping

Correctness, Semantic Equivalence and ReDoS Safety in
LLM-Generated Regular Expressions, and the Limits of Measuring Them

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Abstract

We evaluate eleven contemporary language models on natural-language-to-regex generation over 450 tasks with three samples each (14,850 requests, $10.95), scoring every answer on three axes: whether it satisfies the task's tests, whether it denotes the same language as a human reference, and whether it is vulnerable to regular-expression denial of service (ReDoS).

Our headline result is a negative one about a composite metric we built from the corpus's own metric suite and then had to abandon. A composite requiring correctness, safety and semantic equivalence appears to show that "passing is twice as easy as shipping": models pass 38.0–46.5% of tasks but satisfy all three criteria on only 17.1–23.8%. Decomposing that gap shows that **87% of it is contributed by the semantic-equivalence term**, and a manual audit shows that term is itself dominated by reference-set error and task underspecification rather than model behaviour. Using only the two criteria that never consult a human answer key, the gap is **2.9 percentage points**: just 7.4% of functionally correct patterns are ReDoS-vulnerable.

That figure is itself informative by contrast. Joint correctness-and-security benchmarks in general code generation report far larger penalties. BaxBench finds roughly half of correct backends exploitable. SecureAgentBench reports 15.2% correct-and-secure for its best agent. The regex domain does *not* reproduce that pattern, and we report the difference rather than the similarity. Separately, models are safer than the corpus's own human-written reference patterns — 9.0% pooled against 13.6%, and eight of the eleven separate under an exact McNemar test on the paired tasks. We initially read that as evidence that ReDoS is endemic to human practice. Extending the identical screen to five further populations, including half a million regular expressions extracted from shipped packages, refutes it and replaces it with a sharper ordering. Controlling for task mix by restricting every population, models included, to anchored patterns: those *written to be read* are ReDoS-prone (reusable library entries 20.1%, Stack Overflow answers 17.3%, this corpus's reference answers 13.4%), while those *executed* in shipped packages sit at 8.9%, statistically indistinguishable from the models at 9.8%. *Vulnerability tracks whether a pattern was ever run, not whether a human or a model wrote it.* We calibrate the screen against an independent detector and a dynamic oracle to show the ordering is not an artifact of differential miss rates.

The audit behind this is the paper's methodological contribution. Metrics that compare generated output against a human answer key inherit that key's error rate, a dependency documented for image and text benchmarks [Northcutt et al., 2021] but not, to our knowledge, quantified for this one. In a random sample of disagreements where a model passed every test yet was scored as semantically different, the model was clearly at fault in roughly 15% of cases. The rest split between demonstrably incorrect reference patterns (including a reference character class containing a literal `|`) and prompts that never specified the property in dispute. Because that unreliable term supplies 87% of our composite's headline gap, the composite was measuring its own weakest component. We also show that independent binomial intervals, the default in

this literature, are the wrong inferential tool when all systems answer identical items, and that a paired bootstrap resolves nine times as many pairwise comparisons on the same data.

Replicating the two reference-independent criteria on a second corpus [Ye et al., 2020] shows the qualitative finding holds and the number does not: the same eleven models leave **16.5%** of their correct patterns ReDoS-vulnerable there against 7.4% here, on tasks twenty points *easier*. The correctness-to-security penalty is a property of the corpus, not of the models.

We therefore decline to publish a ranking. We release all raw generations, scoring code and adjudications, and recommend that evaluations separate reference-dependent from reference-independent metrics, assign them different evidentiary weight, and treat any effect size as corpus-local until replicated.

Artifacts. All model outputs, scoring code, per-case adjudications and analysis scripts are available at `https://github.com/plicara/regexeval-2026` [Plicara Labs, 2026b]. Every number in this paper recomputes offline from committed data with no API access.

1 Introduction

Regular expressions make an unusually good probe for language-model evaluation. The task is compact. The output runs. Correctness is partly decidable, which is rare. And a syntactically valid, test-passing answer can still be a security defect, which is what drew us to the task in the first place.

The standard protocol for this task scores a candidate against a reference by DFA equivalence [Locascio et al., 2016, Kushman and Barzilay, 2013], sometimes alongside test-based correctness. Neither criterion notices catastrophic backtracking. A pattern can satisfy every provided example, denote exactly the reference language, and still admit inputs on which matching takes time exponential in the input. That is the ReDoS vulnerability class [Davis et al., 2018], a denial-of-service vector in deployed systems and common enough in human-written code to have prompted ecosystem-scale study.

Siddiq et al. [2024a] closed that gap. Their Re(gEx|DoS)Eval introduces the corpus we use, defines the four metrics we use (PASS@k, VULNERABLE@k, DFA-EQ@k and exact match), scores correctness and security jointly on them, and evaluates T5, Phi-1.5 and GPT-3.5-Turbo. The instrument in this paper is theirs. We say so plainly here because an earlier draft of this work did not, and because it determines what is left for us to contribute: not the measuring apparatus, but what happens when it is turned on the current model population, decomposed, and then turned back on itself.

1.1 Contributions

1. **A decomposition of the joint metric** (Section 4.2). The composite is the natural way to report correctness and security together and it is what the joint-benchmark literature reports (Section 2.1). We show that on this corpus the composite is dominated by its equivalence term, which supplies 87% of the headline gap, and that the term is the least trustworthy of the three. The contribution is the finding that the aggregate is uninformative here, not the aggregate.

2. **A quantified audit of reference-standard error** (Section 5.1). We sample disagreements and adjudicate them, finding that the majority of apparent semantic errors are not model errors. Northcutt et al. [2021] establish that test-set label error is pervasive and can reverse benchmark orderings; we measure it for a corpus whose labels are regular expressions, where a label error is a formal object that can be exhibited and argued with rather than a disputed

annotation. Siddiq et al. [2024a] report their metrics without auditing the reference set they compare against, which is the standard practice we are questioning rather than a defect specific to them.

3. **A human baseline for ReDoS across six populations** (Section 4.4). We apply the identical safety screen to the corpus's own reference patterns, a second NL-to-regex gold set, a grammar-generated control, a reusable-pattern library, Stack Overflow, and half a million regular expressions from shipped packages. The resulting ordering is monotone in whether a pattern is executed rather than in who wrote it, which identifies the mechanism behind the reference set's unreliability instead of only observing it. Screening costs no inference, so this generalises where the model-side results cannot.

4. **An updated model population.** Siddiq et al. [2024a] evaluate T5, Phi-1.5 and GPT-3.5-Turbo. We evaluate eleven models released in the intervening period under pinned serving providers (Section 3.3), and find the population compressed into an eight-point band that a hundredfold price range does not order.

5. **A second-corpus replication** (Section 4.6). We re-run the two reference-independent criteria on StructuredRegex [Ye et al., 2020] with the same models and configuration. The finding replicates in kind and not in size: 16.5% against 7.4%, on easier tasks, with the model ordering not preserved. This is the direct test of Section 4.5's domain-dependence claim, which otherwise rests on comparison across languages and vulnerability classes.

6. **Full artifact release** including raw generations, which permits any of our scoring decisions to be overturned without re-running inference.

1.2 What this paper does not claim

We do not claim the metrics. PASS@k is Chen et al. [2021]; VULNERABLE@k, DFA-EQ@k and exact match on this corpus are Siddiq et al. [2024a]. Joint correctness-and-security scoring is established across several domains (Section 2.1) and on this one. The USABLE composite in Section 3.4 is a conjunction of their three metrics, and the argument of this paper is that it should not be used.

We do not claim the paired bootstrap of Section 5.3. It is Berg-Kirkpatrick et al. [2012], following Koehn [2004], and its use is recommended practice [Dror et al., 2018]. What we report is how much it changes on this data: nine resolvable pairwise comparisons against one under independent intervals. That is an observation about the corpus, not a method.

We do not publish a ranking of the eleven models. Section 5.4 shows that 62% of tasks yield identical outcomes across all eleven systems, and that only nine of fifty-five pairwise comparisons resolve at 95% confidence even under the correct paired test. Reporting an ordinal leaderboard from this data would be unsupported by it.

We do not claim novelty for DFA-equivalence checking, which is standard in this literature [Locascio et al., 2016], nor for ReDoS detection, which has a long static-analysis lineage [Kirrage et al., 2013, Wüstholz et al., 2017, Bhuiyan et al., 2025], nor for the scoring engine, which is prior work by the present authors released separately [Plicara Labs, 2026a] and used here as a pinned dependency.

2 Background and Related Work

NL-to-regex generation. The task was established by Kushman and Barzilay [2013] (KB13) and extended by Locascio et al. [2016] (NL-RX), both of which adopt DFA equivalence against a reference

as the correctness criterion. That is the right choice in principle. `[0-9]+` and `[0-9][0-9]*` denote the same language and differ as strings, so text comparison would mark one of them wrong. But the criterion inherits any defect in the reference set, which is the dependency this paper quantifies.

Re(gEx|DoS)Eval. Siddiq et al. [2024a] supply both the corpus and the metric suite used here. They collect 762 tasks from real user posts, each with a description, positive and negative test strings and a human reference pattern, and score generations on PASS@k, VULNERABLE@k, DFA-EQ@k and exact match, reporting which model produces regexes that are correct *and* secure. Their evaluation covers T5, Phi-1.5 and GPT-3.5-Turbo. Everything this paper measures is measured with their instrument. Our departures are three: we run it on eleven models released since, we decompose the joint metric instead of reporting it whole (Section 4.2), and we turn the safety screen on their reference patterns (Section 4.4), which they do not do. The companion study [Siddiq et al., 2024b] examines ReDoS in LLM-generated patterns alongside developer-forum discussion and reports a skew toward polynomial over exponential vulnerability; we compare against that finding in Section 4.4.

ReDoS. Catastrophic backtracking arises when a backtracking matcher can decompose a prefix in exponentially many ways, canonically via a quantifier applied to a quantified group, as in `(a+)+`. Detecting it statically is a developed area: Kirrage et al. [2013] give an analysis for exponential blow-up by search over the pattern's derivation tree, Wüstholz et al. [2017] extend detection to programs that construct patterns dynamically, and Bhuiyan et al. [2025] survey the detector and mitigation literature. The scoring engine we depend on [Plicara Labs, 2026a] sits in that lineage and combines a structural screen with witness-string execution. On prevalence, Davis et al. [2018] document ReDoS at ecosystem scale and Davis et al. [2019] study regex portability and re-use across languages.

Security of generated code. The observation that models emit working but unsafe code predates the joint benchmarks below. Pearce et al. [2022] generate 1,689 programs across 89 security-relevant scenarios and find roughly 40% vulnerable, and Perry et al. [2023] run a controlled user study showing that participants with an AI assistant wrote less secure code while believing the opposite. Both motivate scoring security separately from correctness rather than assuming the first implies the second.

2.1 Joint correctness-and-security benchmarks

Evaluating functional correctness and security together is an active area, and this paper is not its first instance. Peng et al. [2025] introduce CWEval, scoring 119 security-critical tasks across 31 CWEs and five languages with outcome-driven oracles for both properties, and report a substantial population of functionally correct but insecure generations. Vero et al. [2025] present BaxBench, 392 backend tasks with expert exploits, and find that roughly half of functionally correct solutions are exploitable. Chen et al. [2025] evaluate coding *agents* on 105 repository-level tasks, reporting 15.2% correct-and-secure for the best agent and 9.2% on average. Patir et al. [2025] automate the joint pipeline in DualGauge over 154 tasks and 10 models, observing secure-pass@1 below 12% while functional pass@1 exceeds 50%.

The joint framing is therefore established both across domains and, by Siddiq et al. [2024a], on this one. What differs here is the shape of the instrument. The artifact is a single expression instead of a program. The vulnerability class is ReDoS instead of a CWE taxonomy. Correctness is partly decidable, not established by tests alone. And a human reference exists for every task, which the general-code benchmarks do not have: they execute tests and exploits rather than compare

against a gold artifact. That difference cuts both ways. It makes Section 4.4 possible, and it is the sole reason Section 5.1 is necessary. The reference-free benchmarks are immune to the failure mode we spend half this paper characterising.

Our numbers turned out not to match theirs. Section 4.5 works through the difference.

Sampling-based metrics. We use the unbiased PASS@k estimator of Chen et al. [2021], with the degeneracy noted in Section 3.4.

Significance testing. Comparing systems that answer identical items calls for a paired procedure. The paired bootstrap we use is Berg-Kirkpatrick et al. [2012], following the bootstrap-resampling protocol of Koehn [2004] for machine translation, and its use over unpaired intervals is standard advice [Dror et al., 2018]; the discrete analogue for paired binary outcomes is McNemar [1947]. Section 5.3 reports what the choice costs on this corpus rather than proposing anything new.

Contamination. This corpus was published in 2024 and its tasks come from public forum posts, so it is plausibly inside the training data of every model we evaluate. Sainz et al. [2023] argue that contamination must be measured per benchmark rather than assumed away. We cannot measure it here and say so in Section 6.

Serving-layer variance. Hosted inference routers may serve the same model slug from multiple providers at differing numerical precision. LessWrong [2025] report that 31 of 32 surveyed AI-safety codebases using such a router did not pin the serving provider, making their results a measurement of the router's routing policy in addition to the model. Section 3.3 describes the discipline we adopt in response.

3 Method

3.1 Task and prompt

Each task supplies a natural-language description. The model returns one regular expression. The prompt is fixed across all models and all tasks:

> **system:** You translate a natural-language description into a single Python `re`-compatible regular expression. Reply with ONLY the pattern in one fenced code block, no explanation.
>
> **user:** *{task description, verbatim}*
>
> Reply with only the regular expression pattern in a single fenced code block.

We do not tune this prompt. Prompt optimisation would raise all scores and destroy comparability with prior work on the same corpus. Section 6 notes the corresponding threat.

Extraction. We take the final fenced code block, else the whole trimmed reply, then strip at most one layer of host-language quoting (`r'...'`, `'...'`, `"..."`, backticks, `/.../flags`). Scored literally, `r'\d+$'` is a pattern matching the letter `r`, and would fail for reasons unrelated to regex competence. This affected 7–18 of 1,350 responses per model, under 1.4% throughout, and changes no conclusion. We release both the normalised and the unnormalised scores.

3.2 Corpus

We use Re(gEx|DoS)Eval [Siddiq et al., 2024a], 762 tasks drawn from real user requests, each with a description, positive and negative test strings, and a human-written reference pattern. We evaluate 450 tasks, selected at uniform stride across the corpus rather than as a prefix (the corpus is difficulty-ordered at its head). The subset size was budget-determined.

Match semantics are set per corpus. Under search semantics 761 of the 762 Re(gEx|DoS)Eval references satisfy their own tests, against 94% under full-match. Choose wrong and 46 gold patterns are scored as failures. We verified the loaded configuration by scoring every reference against its own tests before the run (Section 5.1).

The single exception is worth its own sentence, because it is not a mismatch. `regexeval/1660`'s reference, `^((\.)?([a-zA-Z0-9_-]?)(\.)?([a-zA-Z0-9_-]?)(\.)?)+$`, does not accept a string it should reject: it fails to *finish* on one, exceeding the scorer's one-second match timeout, which is counted as a false positive. The safety screen independently flags the same pattern as exponential. The one reference in the corpus that fails its own tests fails because it is ReDoS-vulnerable.

3.3 Models and serving discipline

Eleven models spanning frontier and open-weights systems across a 98× price range. Every request pins a single named provider with fallback disabled, so a request is either served by the named endpoint or fails visibly. Substitution failures did occur and are reported as failures rather than silently re-routed (Section A).

For open-weights models the pin includes quantisation where the router discloses it: GLM-5.2 and DeepSeek-V4-Flash are served at 4-bit by some providers and 8-bit by others, and we pin 8-bit. One pin went stale within an hour of being set (the provider ceased serving that model), so pins are re-verified immediately before each run.

The pin is a request, not a guarantee. So we record which provider actually served each response and check afterwards that it matches. All eleven models came back served entirely by the endpoint they were pinned to.

3.4 Metrics

Let $T = (d, P, N, r)$ be a task with description d, positive examples P, negative examples N and reference pattern r. Let p be a candidate.

Correctness.

$$\text{correct}(p, T) \;=\; [\forall x \in P:\ \text{match}(p, x)] \;\wedge\; [\forall y \in N:\ \neg\,\text{match}(p, y)]$$

evaluated with CPython's `re` under the corpus's declared semantics. This is *reference-independent*.

Semantic equivalence. Both p and r are compiled to finite automata and compared via `dk.brics.automaton` [Møller, 2021], yielding a verdict in $\{\text{EQ}, \text{DIFF}, \text{UNDEC}, \text{UNSUP}\}$. When the verdict is DIFF the engine returns a *witness*: a shortest string in the symmetric difference, which renders the judgment falsifiable by inspection.

Equivalence is decidable only on the regular subset. Backreferences make the language non-regular, and equivalence then has no algorithmic answer at all. Those cases return UNDEC. Between 78 and 133 of each model's scored tasks land there. We therefore report two figures:

$$\text{DFA-EQ} = \Pr[\text{EQ}], \qquad \text{DFA-EQ}_{\text{dec}} = \Pr[\text{EQ} \mid \text{verdict} \neq \text{UNDEC}].$$

The first counts undecidable comparisons as failures. It is a lower bound and cannot flatter. The second isolates the model from the engine's reach. Report only the second and you inflate. Report only the first and you charge the model for a theorem. This metric is *reference-dependent*, which Section 5.1 shows to be decisive.

ReDoS safety. $\text{vuln}(p) = [\text{screen}(p) \neq \textsc{safe}]$, where screen performs structural analysis for known-vulnerable shapes followed by an empirical pass against attack strings under timeout. The structural pass covers three of the five families catalogued by Siddiq et al. [2024b]. So VULNERABLE is a *lower bound*, and SAFE denotes a screening outcome, not a proof. This metric is *reference-independent*. Section 4.4.2 measures how loose the bound is, separately for each population we compare, because a lower bound that is looser in one population than another would produce an ordering by itself.

Where the screen reports a degree, exponential against polynomial, the structural pass infers it from the matched shape and the empirical pass infers it from which probe length first exceeded the timeout. The second is a threshold rather than a measurement of growth order, and we discount the degree split accordingly (Section 4.4.2).

Composite.

$$\text{usable}(p, T) = \text{correct}(p, T) \ \wedge \ \neg\, \text{vuln}(p) \ \wedge \ \left[\text{verdict}(p, r) \neq \textsc{diff}\right]$$

The third conjunct excludes only *proven* difference. Undecidable and unsupported verdicts do not disqualify a pattern. USABLE is reference-dependent through that conjunct, and inherits the error characterised in Section 5.1.

Sampling. With n samples per task of which c satisfy a predicate, we use the unbiased estimator of Chen et al. [2021]:

$$\widehat{\text{metric}}@k \ = \ \mathbb{E}_T \left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}} \right].$$

We draw $n = 3$ and report $k = 3$. At $n = k$ this estimator degenerates to $\mathbf{1}[c \geq 1]$ and buys no variance reduction over the naive any-of-n statistic. We report it as @3 for comparability, not because the estimator is doing any work at this setting. $n > k$ would be required for the estimator's variance properties to apply, and is left to future work with a larger budget.

The estimator is defined for $n \geq k$, and a handful of tasks returned fewer samples than that after a refusal or a spending limit. We exclude those tasks from the affected model's @k estimate rather than score them on what arrived, and report the count in Table 3. Section B records why this is not a cosmetic choice.

3.5 Sampling configuration

Reasoning disabled. Four of the eleven models emit no hidden reasoning tokens. The other seven do. Evaluating them together with reasoning enabled would confound model identity with inference-time compute budget. All requests therefore disable reasoning, and reasoning-token counts are recorded per response to verify compliance (observed: zero throughout).

Temperature unset. We set `require_parameters=true`, which restricts routing to providers that honour submitted sampling parameters rather than silently discarding them. Six of the eleven models reject `temperature` outright. Together, the two settings make the request unroutable. We therefore omit `temperature` entirely and each model samples at its provider default. That keeps the non-silent-discard guarantee and costs us controlled sampling. We think it is the right trade and note it as a limitation.

Since $k > 1$ is only informative under output diversity, we verified diversity directly: repeated queries on a fixed task yielded 3, 3, 3 and 2 distinct patterns for `claude-opus-5`, `gpt-5.6-sol`, `kimi-k3` and `glm-5.2` respectively. The analysis pipeline also warns if $> 90\%$ of a model's tasks return identical samples.

3.6 Controls

Three synthetic responses go through the identical scoring path on every run. The task's own reference must come back correct and usable. The pattern `z{5}` must fail every correctness check. `(a+)+b` must be flagged vulnerable. If any control misbehaves we throw the run away instead of reporting it.

The failure this catches is a scorer that quietly returns nulls, which looks exactly like every model performing badly. All controls behaved on all eleven models.

4 Results

4.1 Main results

Model	n	PASS@3	USABLE@3	95% CI	VULNERABLE@3	DFA-EQ@3	$\text{DFA-EQ}_{\text{dec}}$	EXACT@3	undec.
`kimi-k3`	441	46.5	23.8	[20.0, 27.9]	12.9	14.1	17.1	5.0	78
`qwen3.6-max-preview`	450	42.4	21.6	[17.8, 25.6]	9.1	13.8	17.4	3.8	94
`gpt-5.6-sol`	449	42.1	20.9	[17.1, 24.7]	10.0	9.4	13.3	1.8	133
`claude-opus-5`	432	46.1	20.8	[16.9, 24.8]	13.2	11.6	15.2	2.8	104
`deepseek-v4-flash-0731`	450	38.0	19.8	[16.2, 23.6]	12.0	11.3	14.3	3.3	93
`qwen3.6-plus`	450	39.8	19.8	[16.2, 23.6]	9.8	11.6	14.8	3.8	99
`glm-5.2`	450	42.4	18.7	[15.1, 22.4]	14.2	10.4	12.9	3.8	86
`gpt-5.6-terra`	450	42.2	18.7	[15.1, 22.4]	12.0	10.2	13.1	2.0	100
`gpt-5.6-luna`	449	39.2	18.5	[14.9, 22.0]	11.6	9.1	12.2	1.8	112
`claude-sonnet-5`	450	40.7	18.0	[14.4, 21.6]	10.9	10.4	13.3	3.3	97
`gemini-3.1-flash-lite`	450	38.7	17.1	[13.8, 20.7]	12.0	9.3	11.8	3.1	95

Table 1: Primary results, $n = k = 3$. All figures percentages except n (tasks entering the @3 estimate) and *undec.* (tasks whose equivalence is undecidable). Tasks that returned fewer than three samples are excluded from the estimate rather than scored on what arrived (Section 3.4), which is why n varies; failed requests are itemised in Section A. The interval is a paired bootstrap over tasks (Section 5.3), not an independent binomial one: every model answers the same items, and a binomial interval here would be the estimator this paper spends Section 5.3 arguing against. Ordering is by USABLE@3 and should be read as a banding, not a ranking (Section 5.4) — the intervals overlap almost everywhere, which is the point.

The apparent headline in Table 1 is the gap between PASS@3 (38.0–46.5%) and USABLE@3 (17.1–23.8%): approximately half of all test-passing generations fail at least one of the remaining two criteria, and the ratio is stable across the entire price range.

Section 4.2 shows that this reading is wrong, or at least that it attributes the gap to the wrong cause. We present it here as stated because it is what the composite reports, and because constructing such a composite is a natural thing to do. The point of the next section is that building one without decomposing it was a mistake, and it was ours.

4.2 Decomposing the gap

USABLE conjoins three conditions. Table 2 attributes the drop from PASS@3 to USABLE@3 between the safety term and the equivalence term, and reports alongside it the rate of ReDoS vulnerability *conditional on functional correctness*.

Model	PASS@3	−safety	C&S@3	−equiv.	USABLE@3	vuln.|correct
`kimi-k3`	46.5	2.5	44.0	20.2	23.8	5.6
`qwen3.6-max-preview`	42.4	2.7	39.8	18.2	21.6	7.0
`gpt-5.6-sol`	42.1	1.6	40.5	19.6	20.9	5.6
`claude-opus-5`	46.1	4.9	41.2	20.4	20.8	10.0
`deepseek-v4-flash-0731`	38.0	1.8	36.2	16.4	19.8	5.7
`qwen3.6-plus`	39.8	3.3	36.4	16.7	19.8	8.1
`glm-5.2`	42.4	4.0	38.4	19.8	18.7	9.3
`gpt-5.6-terra`	42.2	2.9	39.3	20.7	18.7	7.3
`gpt-5.6-luna`	39.2	2.9	36.3	17.8	18.5	8.1
`claude-sonnet-5`	40.7	2.7	38.0	20.0	18.0	6.0
`gemini-3.1-flash-lite`	38.7	2.9	35.8	18.7	17.1	8.3
mean	—	2.9	—	18.9	—	7.4
pooled	—	—	—	—	—	**7.4**

Table 2: Decomposition of the PASS→USABLE gap. "C&S" is correct-and-secure: correctness conjoined with safety, omitting the equivalence term. The final column is the share of functionally correct patterns that are ReDoS-vulnerable, and is the one column here computed *per sample* rather than @3: it is a rate over generations, and the @3 form would answer a different question. Denominators are Table 3's restricted ones — tasks with fewer than three samples are excluded — which is why `kimi-k3` reads 46.5 here.

Safety costs 2.9 percentage points on average. Equivalence costs 18.9. So the equivalence term supplies **87%** of the gap that made the composite look interesting.

That split is order-dependent, and the order we chose is the one that favours the other term. Subtracting safety first credits it with every unsafe-correct pattern, including those that would also have failed equivalence, while equivalence keeps only the share no other term had already claimed. Reversing the order can only move attribution toward equivalence. 87% is therefore a *lower bound* on the equivalence term's contribution under any order-symmetric attribution, which is worth saying because the obvious objection to a sequential decomposition is that it is order-dependent, and here the dependence runs against us.

That would be unremarkable if the equivalence term were sound. It is not. Section 5.1 shows, from an independent audit, that roughly 85% of what it counts against a model is reference-set error or an underspecified prompt. Our composite was mostly reporting its own weakest component back to us.

We did not see this coming. The decomposition was run late, after the audit and after the first draft, because a reviewer of an earlier version asked how our numbers compared to joint benchmarks in other domains. Answering that required computing correct-and-secure without the equivalence term, and the answer changed the paper.

Restricting attention to the two reference-independent criteria, the finding is far more modest and far better supported: **7.4% of functionally correct regular expressions are ReDoS-vulnerable** (range 5.6–10.0%).

The denominator matters, so: of the 5,269 generations that satisfied every test, 390 were simultaneously screened as unsafe. That is 7.4%, a rate over *samples* rather than over tasks, which makes it an @1 quantity and the one directly comparable to the single-sample figures in Table 9. At the task level the corresponding count is 144 of 2,051 tasks on which some sample passed. About one in fourteen either way, not one in two. (An earlier draft reported a count of 135 here that we can no longer reproduce from our own released data, and did not state which denominator the 7.4% used.)

EXACT@3, string identity with the reference, runs from 2.0% to 6.3%. String-comparison scoring would therefore misrank essentially every system, which is what justifies the automata-theoretic treatment.

4.3 Sample budget and the fragility of ordering

Because $n = k = 3$ renders the PASS@k estimator degenerate (Section 3.4), we also report @1, which is the per-sample success rate and the quantity most directly comparable to single-sample protocols. Table 3 gives both.

Model	PASS@1	PASS@3	USABLE@1	USABLE@3	VULNERABLE@1	n<3
`kimi-k3`	35.6	46.5	16.0	23.8	9.2	6
`qwen3.6-max-preview`	37.0	42.4	16.5	21.6	7.4	0
`gpt-5.6-sol`	37.3	42.1	16.0	20.9	8.4	1
`claude-opus-5`	41.4	46.1	17.3	20.8	10.7	12
`deepseek-v4-flash-0731`	30.0	38.0	14.4	19.8	7.7	0
`qwen3.6-plus`	35.5	39.8	15.9	19.8	8.0	0
`glm-5.2`	34.1	42.4	13.5	18.7	9.6	0
`gpt-5.6-terra`	36.6	42.2	14.8	18.7	8.9	0
`gpt-5.6-luna`	33.8	39.2	15.2	18.5	8.7	1
`claude-sonnet-5`	36.8	40.7	15.3	18.0	9.3	0
`gemini-3.1-flash-lite`	33.2	38.7	13.9	17.1	9.5	0

Table 3: Per-sample (@1) against any-of-three (@3) estimates, computed from per-sample success counts. The final column gives the number of tasks with fewer than three usable samples, which are excluded from that model's @3 estimate. The @3 columns agree with Table 1 to the digit for all eleven models, which is the check that matters: they are computed by different routes over the same exclusion, and they used to disagree by up to 1.3 points. Reconciling them is how the estimator defect in Section B was found.

The comparison is instructive beyond bookkeeping. At @3, `kimi-k3` attains the highest USABLE at 23.8%. At @1 the leader is `claude-opus-5` on 17.3%, and `kimi-k3` drops to joint third. *The induced ordering depends on the sample budget.* Systems differ not only in per-attempt quality but in how much they benefit from additional attempts, and a leaderboard that fixes k silently fixes a weighting between those two properties. This is a further reason to prefer banded reporting (Section 5.4), and a reason to state k prominently whenever an @k metric is ranked.

4.4 ReDoS and the human baseline

Table 4 applies the identical safety screen to the corpus's 450 human-written reference patterns. For comparability with the human set (one pattern per task) we score one sample per task per model rather than the any-of-three statistic of Table 1.

Source	n	vulnerable (%)	exponential (%)	polynomial (%)	McNemar p
Human reference answers	450	13.6	6.4	7.1	—
`kimi-k3`	447	10.7	5.4	5.4	0.136
`claude-opus-5`	444	10.6	7.4	3.2	0.065
`glm-5.2`	450	10.2	5.8	4.4	0.058
`gemini-3.1-flash-lite`	450	9.8	5.1	4.7	0.033
`gpt-5.6-sol`	450	9.3	5.1	4.2	0.018
`claude-sonnet-5`	450	8.9	6.0	2.9	0.005
`gpt-5.6-terra`	450	8.7	4.9	3.8	0.004
`qwen3.6-plus`	450	8.4	5.1	3.3	0.001
`gpt-5.6-luna`	450	8.0	4.0	4.0	0.001
`deepseek-v4-flash-0731`	450	7.3	4.4	2.9	0.000
`qwen3.6-max-preview`	450	7.3	4.7	2.7	0.000
All models pooled	4941	9.0	5.3	3.8	—

Table 4: ReDoS screening of model outputs against the corpus's own human-written reference patterns, one pattern per task in all cases. The final column is the exact McNemar test against the reference set on the same tasks; see below for why an independent interval would be the wrong instrument here.

Every model screens safer than the reference set, and eight of the eleven do so distinguishably. That second clause is the one that needs the work, and an earlier draft did not do it: it read the ordering off eleven independent binomial intervals over a pooled 4,941 patterns. That is the wrong estimator twice over, and both errors are ones this paper criticises elsewhere. The 4,941 are eleven models answering the same 450 tasks, so task difficulty is a shared random effect and a pooled binomial interval is too narrow (Section 5.3). And each model is compared against the reference set *on the same tasks*, so the comparison is paired and an unpaired interval discards the pairing — with $n \approx 450$ a side and a three-point difference, nothing would resolve.

McNemar's test is the paired procedure for this design, and we use the exact binomial form because the discordant counts are small (47 to 65 per model). Everything the model and the reference have in common cancels. Against `qwen3.6-max-preview`, for instance, the reference set is vulnerable where the model is safe on 39 tasks and safe where the model is vulnerable on 11 ($p = 0.0001$). Against `kimi-k3` the split is 39 to 26 ($p = 0.14$), which does not resolve. The three models that do not separate — `kimi-k3`, `claude-opus-5` and `glm-5.2` — are the three with the highest vulnerability rates, which is what one would expect and is not what an eleven-way ordering of point estimates would have told us.

So: models are safer than this corpus's answer key, on most of the field, under the right test. Read alone, that invites a strong reading: ReDoS vulnerability is endemic to human-written regular expressions, and models merely reproduce it at a lower rate. We drafted that claim. It does not survive the obvious check, which is that a benchmark answer key is not the same population as working code.

4.4.1 Six populations

Screening a pattern requires no inference, so the human side of this comparison can be extended to any corpus at no cost. We applied the identical screen to five further populations: the gold answers of KB13 [Kushman and Barzilay, 2013]; the targets of NL-RX [Locascio et al., 2016], generated from a grammar rather than written by people and included as a control; and three corpora from the artifact of Davis et al. [2019], namely regular expressions extracted from shipped packages across eight registries, regular expressions posted to Stack Overflow, and the reusable patterns published on `regexlib.com`. Samples are drawn at a fixed seed.

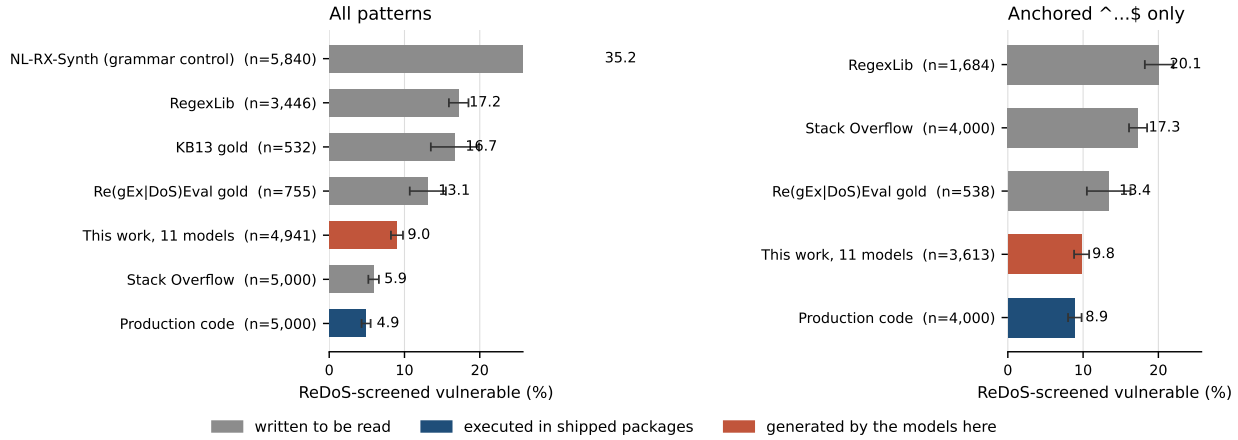

Figure 1: The same screen on every population, before and after controlling for task mix. Bars are 95% binomial intervals. The restriction is the same rule on both sides — keep only anchored `^...$` patterns — and it is applied to the models too, which an earlier draft did not do. What survives it is not an ordering of humans against machines.

Production code screens at 4.9%, below every model we evaluated. The endemic reading is therefore wrong, and we would have published it.

The raw comparison is confounded by task mix. This corpus is unusually rich in validators (email, ISBN, hostname), the construction that backtracks, whereas most regular expressions in the wild are short fragments with no opportunity to. Restricting every population to anchored `^...$` patterns gives the closest matched populations available, and the ordering that emerges does not follow the axis we expected:

> The dividing line is not human against machine. It is whether the pattern was ever run. Every population written to be *read* — library entries at 20.1%, forum answers at 17.3%, benchmark reference answers at 13.4% — is ReDoS-prone. The one population that has been *executed* under real traffic sits at 8.9%, and the models sit with it at 9.8%.

The model row is restricted the same way. An earlier draft of this table put the models into the anchored block at their *unrestricted* rate, taken straight from Table 4. That compares an anchoring-restricted human population against an unrestricted model one, which is precisely the confound the restriction exists to remove, and a reviewer was right to refuse it. The row now reports the models' rate on their own anchored outputs — the identical rule, applied to the pattern rather than to its provenance — which is 9.8% ± 1.0 over 3,613 patterns.

Population	Written to be	n	vulnerable
All patterns			
NL-RX-Synth (grammar-generated control)	—	5,840	35.2% ± 1.2
RegexLib, published for reuse	read	3,446	17.2% ± 1.3
KB13 gold answers	read	532	16.7% ± 3.2
Re(gEx|DoS)Eval gold answers	read	755	13.1% ± 2.4
Stack Overflow posts	read	5,000	5.9% ± 0.7
Production code	run	5,000	4.9% ± 0.6
Anchored `^...$` only, controlling for task mix			
RegexLib, published for reuse	read	1,684	20.1% ± 1.9
Stack Overflow posts	read	4,000	17.3% ± 1.2
Re(gEx|DoS)Eval gold answers	read	538	13.4% ± 2.9
This work, 11 models pooled	—	3,613	**9.8% ± 1.0**
Production code	**run**	4,000	**8.9% ± 0.9**

Table 5: The identical ReDoS screen applied to six populations, plus this paper's models under the identical restriction. Intervals are 95% binomial, which is the right form here and not in Table 1: these populations are genuinely independent — different corpora, no shared items — where the models answer shared tasks and need a paired procedure. The anchored block is the comparison to read: the unanchored figures are dominated by task mix, which is why Stack Overflow moves from 5.9% to 17.3% between the two blocks.

Correcting it moves the models *toward* the answer key and away from production code, by about eight tenths of a point, and the conclusion survives anyway: 9.8% against production code's 8.9% is not a resolvable difference ($z = 1.27$, $p = 0.20$), while both sit far below every population written to be read. We report the direction of the correction because it cost us something.

Two stricter matchings are available here that are not available for the wild populations, because we know which *task* each model pattern answers. Restricting to the 330 of 450 tasks whose reference is itself anchored, the models screen at 10.6% against the reference set's 13.9%. Requiring both — an anchored output on an anchored task — gives 11.4% against 13.9%, and at that point the model-versus-reference difference is no longer resolvable either. That is the honest ceiling on how strongly the reference comparison can be stated, and it does not disturb the production comparison, which is a population-level restriction applied identically on both sides.

This is a measurement of the mechanism rather than a conjecture about it. A pattern published for reuse, posted in an answer, or written to key a benchmark is authored once to communicate an idea, and nothing subsequently happens to it. A pattern in a shipped package is executed, profiled, and occasionally repaired; Davis et al. [2018] document exactly that remediation activity at ecosystem scale. Vulnerability tracks exposure to execution, and answer keys have none.

A competing explanation produces the same ordering without any repair. Packages under a lint rule such as `safe-regex` never contained the bad pattern to begin with, so selection at authoring time would look exactly like survival after it. The two are separable in principle — repair leaves a commit and prevention does not — and we cannot separate them here: the artifact we screen carries registry and module counts but no maintenance history, so we can stratify by how widely a pattern is used and not by whether its package is still maintained. Davis et al.'s documented remediation supports the repair mechanism and does not rule out the other. We state the finding as exposure-to-execution because that is what both mechanisms share, and flag the decomposition as open.

It also explains why this corpus's answers are ReDoS-prone without appeal to authorial careless-

ness. Re(gEx|DoS)Eval was assembled from real user posts, and forum posts screen at 17.3%. That the gold set is safer than its own source population (13.4%) suggests its authors curated, which is to their credit and does not close the gap to executed code.

The control is the clearest case of the mechanism. NL-RX-Synth screens at 35.2%, more than double the next population and seven times production code. That is not the screen misbehaving on an odd construct distribution, and the reasons say so: every one of the 2,057 vulnerable patterns is flagged by the *structural* pass, none by the empirical one, and they divide into 994 with a quantifier wrapping a quantified group, 891 with two quantifiers in sequence over overlapping character sets, and 172 with a quantifier over overlapping alternation. Representative targets are `.*(.*)([a-z]).*`, `(dog.*[0-9].*)+` and `((dog)|(..*[AEIOUaeiou].*)){4,}`.

The generating grammar composes `.*`, `(...)*` and `(...){n,}` freely over overlapping character classes, and those compositions *are* the backtracking shapes. A grammar that samples uniformly from a space of regular expressions has no reason not to, because nothing downstream of it ever runs the output. Far from being an outlier that impugns the screen, NL-RX-Synth is the purest instance of this section's mechanism: it is the population with the least exposure to execution of any we measured, and it is the most vulnerable. Section 4.6 returns to this, because StructuredRegex is grammar-built too.

Section 5.1 reaches the same conclusion from semantics, and this section reaches it from safety without consulting the reference at all: the reference set is the least reliable artifact in the evaluation. The revised finding is weaker than the one we drafted, better supported, and preserves the practical implication. Safety screening must be applied at the point of use, because neither the generator, nor the reference author, nor the benchmark performs it.

4.4.2 Is the screen equally sensitive across these populations?

The ordering above is only a fact about the populations if the screen misses vulnerable patterns at the same rate in all of them. If production patterns are longer and structurally unlike RegexLib's showcase validators, a differential miss rate could generate the entire result. "VULNERABLE is a lower bound" is an adequate caveat for one population and is not one for six, so we measure it.

The screen does not see every dialect. Every population is screened by CPython's `re`, and patterns it will not compile are dropped. That filter is not uniform, which is the part that matters: it removes 5.0% of the production pool (26,900 of 537,804) against 11.4% of Stack Overflow and 10.2% of RegexLib, and within production it runs from `cpan` at 7.7% down to `pypi` at 0.3%. Section 6 applies exactly this scrutiny to KB13 and NL-RX, whose DSL `re` accepts as literals, and the corpus doing the most work in this paper did not previously get it.

What is removed is `\A...\z`, `\G`, `\Q...\E`, Perl's `\x{...}` and Unicode properties — Perl and Ruby syntax, and it is enriched for the anchored validator shapes the matched comparison rests on: 23.7% of what is dropped is anchored against 17.2% of what is kept. We had assumed this biased in our favour and it does not obviously do so: the read-to-be-read populations lose the larger share, and losing vulnerable patterns from those would understate the very gap we report. Assuming either direction is guessing, so we measure it. We translate what can be translated without guessing (`runner/dialect.py`: `\A`→`^`, `\z`→`\Z`, `\Q...\E` to an escaped literal, named groups to Python's spelling), decline what cannot (`\G` anchors to the previous match and has no Python equivalent; `\p{...}` would have to be approximated), and report the recovered population as a robustness row alongside the original.

Two registries cannot backtrack at all. `crates.io` and `godoc` target RE2 and the Rust `regex` crate, which are linear-time by construction. ReDoS is not a property their patterns can have, whatever their shape, so screening them as though they ran under a backtracking engine is a category error. 23,941 patterns appear in one of the two; of those, 20,658 appear in no backtracking registry at all, and those are the ones the robustness run excludes. (A pattern can be published to several registries, so this is a count of distinct patterns rather than the sum of the two registry columns.)

Neither correction moves the number. Normalisation recovers 17,151 of the 26,900 dropped production patterns (64%). What stays out is mostly not exotic syntax: the largest group, 4,588 patterns, is malformed or truncated by the static extraction that produced the corpus — template fragments rather than regular expressions — followed by `\G` and Unicode properties, which we decline rather than approximate. Screening the recovered population takes anchored production from 8.9% to 9.1%; dropping the linear-engine registries takes it to 8.3%; doing both gives 8.2%. The whole range is 8.2–9.1%, well inside the interval on any one of them, and the models at 9.8% remain indistinguishable from every variant. The dialect filter and the engine mismatch are real defects in the instrument, and they are not what produces the result.

Recall, measured per population. Agreement between two static approximations is not calibration, so we pair an independent detector with a dynamic oracle. The detector is `weideman-RegexStaticAnalysis`, from the artifact of Davis et al. [2019] and the same one their ecosystem study used: a different analysis, a different language, a different author. When it calls a pattern vulnerable it emits an exploit string, and we then build that input at growing pump counts and time *CPython's own matcher* on it under a hard timeout. A pattern is ground-truth vulnerable when matching either fails to finish or grows measurably super-linearly in the input length. That is a fact about the engine this paper screens with, not a second opinion.

Population	sampled	unanalysable	caught/confirmed	recall (%)	95% CI	agreement (%)	med. len
Stack Overflow	240	80	25/27	92.6	[77, 98]	67.9	27
RegexLib	240	89	33/36	91.7	[78, 97]	65.0	67
Re(gEx|DoS)Eval gold	219	60	19/22	86.4	[67, 95]	65.8	46
Production code	240	46	16/20	80.0	[58, 92]	74.2	23
NL-RX-Synth	240	56	16/21	76.2	[55, 89]	62.1	24
Model outputs	240	64	11/15	73.3	[48, 89]	55.4	55
KB13	209	110	2/3	66.7	[21, 94]	52.6	15

Table 6: Screen sensitivity per population. *Unanalysable* counts patterns the independent detector skipped or timed out on, which are neither evidence for nor against. *Caught/confirmed* is our screen's hits over the patterns the detector flagged *and* a timing measurement then confirmed blow up under CPython. Recall is measured against what the detector finds, so it is a relative sensitivity rather than an absolute one — which is what the cross-population comparison needs, since the question is whether the instrument is equally blind everywhere and not how blind it is. The intervals are wide because dynamically confirmed vulnerabilities are scarce in a sample of a few hundred; they are given so that the comparison is read at the precision the counts support.

Recall runs 67–93% across the seven populations, and the intervals overlap almost everywhere: dynamically confirmed vulnerabilities are scarce in a sample of a few hundred, and KB13 yields three. So this does not resolve a fine ordering of sensitivities, and we do not claim one.

What it does resolve is the direction, which is the part that mattered. If the ordering were manufactured by the screen being blinder in production code than in showcase validators, recall would have to be *lower* in production. It is not. The two populations where the screen is most sensitive are `regexlib.com` at 91.7% and Stack Overflow at 92.6% — the two with the highest measured vulnerability — while production code sits at 80.0% and this corpus's gold answers at 86.4%. The differential runs the opposite way to the one that would explain away the result.

Dividing each measured rate by the recall behind it gives a first-order sensitivity check. On the anchored populations that turns 20.1%, 17.3%, 13.4%, 9.8% and 8.9% into 21.9%, 18.7%, 15.5%, 13.4% and 11.1%. Every rate rises, because the screen is a lower bound everywhere; the *ordering is unchanged*; and the read-against-run gap is not closed. We would not put those corrected numbers in a table — recall here is measured against what one detector finds, so it is relative rather than absolute, and the counts behind it are small — but the check is the one a reader should want, and it does not overturn anything.

Two limitations belong with it. The independent detector could not analyse between 46 and 110 of each 240-pattern sample, and those are neither evidence for nor against; ground truth therefore covers the patterns two tools can both reason about, which is not all of them. And agreement between our screen and the detector runs 53–74%, far below either one's recall, because the detector flags a good deal that does not blow up in CPython — `^[a-z]+[a-z]*$` is quadratic in principle and instant in practice. That gap is the reason this section uses a timing measurement as the arbiter rather than a second opinion.

Relation to prior work, and a caveat on the split. Siddiq et al. [2024b] report that LLM-generated regexes skew toward polynomial ReDoS. We do not reproduce this in our models: pooled, we observe 5.3% exponential against 3.8% polynomial. The cross-corpus run locates the skew elsewhere. In the anchored production sample, polynomial exceeds exponential 253 to 105, while this corpus's gold answers are near even at 34 to 38.

That comparison is weaker than it looks and we would rather say so than let it be read as a finding. Our exponential-versus-polynomial split is only partly a claim about growth order. The structural pass assigns it from the shape it matched, which is principled; the empirical pass assigns it from *which probe length first timed out*, which is a threshold, not a measurement. Patterns flagged only empirically therefore carry a degree label our instrument did not really establish.

How much of the split that affects, we can say for one population. Among the generations that satisfied every test, on both corpora, *every* vulnerable verdict came from the structural pass and none from the empirical one, so for those the degree is shape-derived rather than threshold-derived. We have not established the same for the reference sets or for the wild populations, where patterns are longer and more of them fall outside the structural pass's grammar. The subcategory counts are also small. We report the discrepancy with prior work as an observation that wants a replication under a detector that measures growth order directly, and draw nothing from it.

4.5 Comparison against joint benchmarks in other domains

Because USABLE carries a semantic-equivalence conjunct that the joint benchmarks of Section 2.1 do not, comparing it directly to their figures would understate us. We therefore also report *correct-and-secure*: correctness conjoined with safety, omitting the equivalence term, which is the construction those benchmarks use.

We do not reproduce the pattern. Vero et al. [2025] report that roughly half of functionally correct backends are exploitable. Chen et al. [2025] report 15.2% correct-and-secure for their strongest agent, against substantially higher functional correctness. Patir et al. [2025] report

Benchmark	tasks	functional (%)	joint (%)	vuln.|correct (%)
This work (regex, ReDoS)	441	42	39	7
BaxBench [Vero et al., 2025]	392	62 (best)	—	≈50
SecureAgentBench [Chen et al., 2025]	105	—	15.2 (best), 9.2 (mean)	—
DualGauge [Patir et al., 2025]	154	>50	<12	—

Table 7: Our correct-and-secure rate against joint correctness-security results from other domains. Task types, vulnerability classes and evaluation methods all differ substantially. The comparison is one of magnitude and of the ratio between functional and joint success, not of models or difficulty.

secure-pass@1 below 12% while functional pass@1 exceeds 50%. In each case the security criterion removes a large majority of functionally correct output.

In our setting it removes 7.4%.

The one prior correct-and-secure figure on *this* corpus is Siddiq et al. [2024a]'s, over T5, Phi-1.5 and GPT-3.5-Turbo. We do not put it in Table 7. Their generation, extraction and dialect handling differ from ours in ways we did not control for, so a difference between their number and ours would not be attributable to the two model populations, which is the only comparison that would be interesting. Anyone wanting that comparison should re-run their harness on the current models rather than read it off the two papers.

We have several explanations for the divergence. One of them is testable on this data and the test is in Section 4.6; the other two are not, and we say which is which rather than listing all three as equally open:

1. **Attack surface.** A regular expression is a single expression with one failure mode of interest. A backend application has many independently exploitable components, so the probability that at least one is vulnerable compounds with program size.

2. **Vulnerability class.** ReDoS is a structural property of the pattern, plausibly well represented in training data as a recognised anti-pattern. CWE-classified defects in application code are more diverse and more context-dependent.

3. **Task difficulty ceiling.** Our functional pass rate is itself low (38–47%), so the correct subset may be biased toward simpler tasks whose solutions have less structural room for catastrophic backtracking.

The third is the one that would deflate the finding, and it is the one we can test, because we have a second corpus that is twenty points easier. If the correct subset is safe because it is simple, then StructuredRegex's correct patterns should be simpler still and therefore safer. They are neither. On the structural measures, StructuredRegex's correct patterns are *longer* (median 33 characters against 28) and carry *more* quantifiers (median 4 against 3), while being twice as ReDoS-prone. Section 4.6 takes this further, because the direction of the difference turns out to be informative about which shape is doing the damage.

The first two we cannot separate here. Both predict that the regex domain looks safer than backend code, and neither predicts anything about the gap between our two corpora, so nothing in this study distinguishes them.

So the claim that generalises is weaker than "correct code is often insecure," and more useful. *The size of the correctness-to-security penalty depends on the domain and has to be measured there.* Assuming it transfers is what we did, and we were wrong. Worse, our composite showed a gap of about the expected size for entirely the wrong reason (Section 4.2). An undecomposed joint metric

will mislead you exactly there, where its aggregate agrees with the literature and its components do not.

4.6 Replication on a second corpus

Every figure above comes from one corpus, so the 7.4% is a property of Re(gEx|DoS)Eval until shown otherwise. The two criteria that never consult a reference can be computed anywhere that supplies test strings, and StructuredRegex [Ye et al., 2020] supplies them: each instance carries positive and negative example strings alongside its description. Its own targets are in a prefix DSL and we never read them, so this run has no DFA-EQ and no EXACT term, which suits the argument of this paper.

We collected one sample per task from the same eleven pinned models under the same configuration, 622 of the 629 test instances (seven ship no positive or negative strings, leaving PASS@k undefined), for $3.67. Table 8 reports the 513 tasks every model answered. Section B records why that number is not 622.

Model	PASS@1	95% CI	VULNERABLE@1	correct-and-secure	VULNERABLE |correct
`claude-sonnet-5`	66.1	[61.9, 70.0]	14.2	56.3	14.7
`gpt-5.6-sol`	65.5	[61.3, 69.5]	14.0	55.8	14.9
`claude-opus-5`	61.6	[57.3, 65.7]	14.4	51.9	15.8
`gpt-5.6-terra`	62.2	[57.9, 66.3]	16.0	51.9	16.6
`qwen3.6-max-preview`	60.6	[56.3, 64.8]	13.6	51.7	14.8
`gpt-5.6-luna`	59.8	[55.5, 64.0]	15.0	50.1	16.3
`gemini-3.1-flash-lite`	60.0	[55.7, 64.2]	16.6	48.7	18.8
`kimi-k3`	58.9	[54.6, 63.0]	17.2	48.7	17.2
`qwen3.6-plus`	55.9	[51.6, 60.2]	15.6	46.4	17.1
`glm-5.2`	55.9	[51.6, 60.2]	17.9	45.0	19.5
`deepseek-v4-flash-0731`	51.7	[47.3, 56.0]	15.8	43.5	15.8
Pooled	59.8	—	15.5	50.0	**16.5**

Table 8: StructuredRegex, one sample per task, on the 513 instances answered by all eleven models. Percentages. Intervals are Wilson binomial, which is appropriate here in a way it is not in Table 1: this is one sample per task, so there is no within-task resampling for a bootstrap to exploit. The last column is the share of functionally correct patterns that are ReDoS-vulnerable, the same quantity this paper reports as 7.4% on Re(gEx|DoS)Eval.

	Re(gEx|DoS)Eval	StructuredRegex
Tasks	441	513
Reference used in scoring	yes (DFA-EQ)	no
PASS@1, range	30.0–41.4%	51.7–66.1%
VULNERABLE@1, range	7.4–10.7%	13.6–17.9%
VULNERABLE |correct, pooled	**7.4%**	**16.5%**

Table 9: The two reference-independent criteria across both corpora, at @1 in each so the comparison is like for like.

The qualitative result replicates and the quantitative one does not. Working patterns are exploitable at a material rate on both corpora, which is the finding. But the rate is **16.5%** here

against 7.4% on Re(gEx|DoS)Eval, and the obvious deflationary reading, that harder problems produce worse patterns, is ruled out by the direction of the difficulty gap: StructuredRegex is *easier*, by roughly twenty points of PASS@1.

Which shape, and why a grammar produces it. The structure of the correct patterns says where the extra vulnerability comes from, and it is not where one would guess. StructuredRegex's correct patterns are longer (median 33 characters against 28) and carry more quantifiers (median 4 against 3), but they are *less* nested: a quantifier applied to a group — the `(a+)+` shape everyone pictures when they hear ReDoS — appears in 18.5% of them against 34.7% on Re(gEx|DoS)Eval, and their median group nesting depth is zero. The corpus with fewer of the canonical exponential shapes is the corpus with twice the vulnerability rate.

What it has instead is repetition in *sequence*, and the screen's own reasons say so. Of the vulnerable correct patterns on Re(gEx|DoS)Eval, 236 are flagged for a quantifier wrapping a quantified group and 154 for two quantifiers in sequence over overlapping character sets — 40% adjacent. On StructuredRegex the shares invert: 202 nested against 355 adjacent, 64% adjacent. In absolute terms the nested count barely moves between corpora while the adjacent count more than doubles. The extra vulnerability is entirely the polynomial family: `\s*\s*`, `[a-z]+[a-z0-9]*`.

Which is what its targets are built out of. A grammar over repetition, optionality and concatenation produces adjacency by construction, and Section 4.4.1 found the identical signature in NL-RX-Synth, the other grammar-built corpus here: 891 of its 2,057 vulnerable targets are flagged for that same shape, and the population screens at 35.2%. Two grammar-generated corpora, built by different authors for different purposes, over-represent the same family.

We think the reading is this. A grammar samples compositions uniformly and has no reason to avoid adjacency, because nothing downstream of it runs the output; a human writing a validator writes one nested quantifier and stops. So the correctness-to-security penalty depends not only on the domain but on how the corpus's targets were *authored*, which is a sharper and more uncomfortable claim than the one we set out with.

Section 4.5 argues from BaxBench and SecureAgentBench that the correctness-to-security penalty is domain-dependent. That argument compares across languages and vulnerability classes, and a reader is entitled to discount it. The same conclusion now holds between two corpora of the same task, in the same language, under the same screen and the same eleven models, with the effect size differing by a factor of 2.2. We regard this as the stronger form of the claim, and it cuts against our own headline number.

The ordering of models is also not preserved. `claude-sonnet-5` leads here and sits mid-table on Re(gEx|DoS)Eval; `kimi-k3` leads there and sits eighth here. Section 5.4 declines to publish a ranking on single-corpus evidence, and this is the direct test of that caution.

4.7 Cost

The extremes differ by a factor of 98 in price and 1.1 percentage points in USABLE@3 — and in the direction that favours the cheaper model, which scores 19.8% against 20.8%. That difference is well inside what we can resolve (Section 5.3), which is to say the two are indistinguishable. This observation is about this task only. It is not a general claim about model capability.

4.8 Refusals

A content filter blocked `claude-opus-5` on 29 requests, 2.1% of its calls, spread across five tasks. Four have some security flavour: strings that exclude a single quotation mark, a six-character

Model	USABLE@3 (%)	cost/request ($\$\times 10^{-6}$)	total ($)
`deepseek-v4-flash-0731`	19.8	25.6	0.03
`gpt-5.6-luna`	18.5	42.9	0.06
`gemini-3.1-flash-lite`	17.1	90.2	0.12
`qwen3.6-plus`	19.8	120.6	0.16
`glm-5.2`	18.7	158.2	0.21
`qwen3.6-max-preview`	21.6	387.9	0.52
`gpt-5.6-terra`	18.7	406.2	0.55
`claude-sonnet-5`	18.0	932.4	1.26
`kimi-k3`	23.8	1328.1	1.79
`gpt-5.6-sol`	20.9	2107.6	2.85
`claude-opus-5`	20.8	2514.2	3.39

Table 10: Cost per *request*, measured from per-request billing returned by the serving layer rather than estimated from list prices. A task costs three of these at $n = 3$: `claude-opus-5` at 2514.2×10^{-6} over 1,350 requests gives the $3.39 total.

alphanumeric password, hex byte sequences, and SQL update and insert statements. The fifth is *"Matches a file extention."* No other model refused anything.

The refusals were not deterministic. Several tasks came back refused on one sample and answered on the other two, under identical settings. You can only see that at $k > 1$. A single-sample protocol would have logged a flat failure and moved on. We count refusals as their own failure category, not as wrong answers (Section A).

4.9 Inference-time compute

On the corpus's simplest task (*"Matches exactly 1 numeric digit (0-9)."*), `qwen3.6-max-preview` emitted 1,571 hidden reasoning tokens before answering `^[0-9]$`, at a cost of $0.0098. With reasoning disabled it produced the identical answer in 10 tokens at 1/100th the cost. Asking that model for *low* reasoning effort produced *2,375* reasoning tokens, more than the default. Effort controls are not reliably monotone.

A reasoning-enabled comparison on 12 tasks across five models yielded per-request costs 15.7× higher and no interpretable accuracy signal at that sample size (±14 percentage points per cell). We report the cost result and explicitly decline to report an accuracy conclusion.

5 Measurement Validity

5.1 The reference standard contains errors

A benchmark's answer key is a set of labels, and labels have an error rate. Northcutt et al. [2021] audit ten widely used test sets and find a mean 3.3% error rate, enough in several cases to reverse the ordering of models the benchmark was being used to compare. The corpus here differs in a way that helps: a label is a regular expression, so a disputed label can be settled by exhibiting a string on which the reference and the description disagree, rather than by a second opinion. Every adjudication below rests on such a string.

First we ruled out a loading fault. 449 of 450 reference patterns satisfy their own tests, so the corpus semantics and dialect are configured correctly. The real defect is quieter. A reference can pass its own tests and still fail to denote what its description says, whenever the tests never exercise the difference.

We drew a seeded random sample of 60 cases in which a model *passed every test* yet received a DIFF verdict, and adjudicated 14 in detail. The 14 are *the first 14 in seed order whose witness string is ASCII*. That rule matters for the arithmetic below, so it is stated rather than left to be recovered from the released sample: the non-ASCII cases are a known and separate artifact, and adjudicating them would have been adjudicating Python's Unicode defaults rather than either party's regular expression.

Adjudicated cause	count	share
Reference pattern is incorrect	5	36%
Description underspecifies the disputed property	6	43%
Model output is incorrect	3	21%

Table 11: Adjudication of 14 sampled DIFF verdicts on test-passing outputs. Per-case reasoning is released.

Separately, 19 of the 60 sampled disagreements (32%) differ only on non-ASCII input. Python's `\d` is Unicode-aware and `[0-9]` is not, so the two are formally distinct languages and practically the same thing.

Because the adjudicated 14 were drawn from the ASCII-witness cases only, none of them is among those 19, and the two effects compose without double-counting: the model is at fault in 3 of 14 ASCII-witness cases, and 41 of the 60 sampled cases have an ASCII witness, so the model is clearly at fault in $(41/60) \times (3/14) = 14.6\%$ of DIFF verdicts. Call it roughly **15%**. The verdict labels in Table 11 map to the released adjudications as `model_right` → reference incorrect, `ambiguous` → description underspecifies, `gold_right` → model incorrect.

Representative cases:

- **Description:** *"It just accepts only positive numbers."* The reference `^\d+([.,]?\d+)?$` admits `0`. The model excluded zero and was scored incorrect.
- **Description:** *"A very simple ISBN validation expression—it just checks for a 10 digit number."* The reference is `^\d{9}[\d|X]$`. The character class contains digit, *pipe* and `X`: an alternation operator written inside a class, where it denotes a literal. The reference therefore accepts `000000000|`, which is the witness separating it from the model's `^\d{9}[\dX]$`.
- **Description:** *"... make sure commas are in the rite place (if present)."* The reference

  ```
  ^\$?\d{1,3}(,?\d{3})*(\.\d{1,2})?$
  ```

  makes the comma optional, admitting `0,000000` and negating the stated purpose. The model enforced comma placement and was scored incorrect.
- **Underspecification:** *"Matches any single upper- or lower-case letter."* The reference `^[a-zA-Z]$` requires a whole-string match. A candidate `[A-Za-z]` requires only occurrence. The description does not distinguish the two. We tested whether anchoring alone explains failures generally: it accounts for approximately 6%, so the effect is diffuse rather than a single correctable convention.

Consequence. DFA-EQ and USABLE are lower bounds on model correctness by an unmodelled margin. PASS and VULNERABLE are unaffected, since neither consults the reference.

Limitation of this analysis. Fourteen cases, judged by us, on our own benchmark. That is a weak basis for a precise number and we do not want it cited as one. We release the reasoning for each case, and the unadjudicated cases keep an empty verdict field so someone else can fill them in. Anyone quoting 15% should re-derive it from a larger sample with annotators who have no stake in the answer. We report it because the direction is not in doubt, and because the direction is what changes how the composite should be read.

5.2 The composite rewards putting an answer beyond checking

DFA-EQ counts an UNDEC verdict as a failure. USABLE does the opposite: its third conjunct excludes only a *proven* DIFF, so a comparison the engine cannot answer is scored as a pass. Each convention is defensible alone — one is a lower bound that cannot flatter, the other isolates the model from the engine's reach — but together they create an incentive. Equivalence is undecidable exactly when a pattern uses backreferences, so a model that reaches for one is scored *more* usable for having put its answer beyond checking. Between 78 and 133 tasks per model land there, a spread of 55, which is large enough to matter.

The obvious test is to correlate a model's UNDEC count against its USABLE@3. That returns nothing (Pearson -0.13), and it returns nothing misleadingly: undecidability credit and equivalence skill push the composite in opposite directions, and a correlation on the sum nets them out. So we decompose instead. Table 12 recomputes USABLE@3 under the stricter rule that the verdict must be *proven* EQ.

Model	undec.	USABLE@3	proven-EQ only	tasks	share of USABLE (%)
`kimi-k3`	78	23.5	13.0	47	44.8
`qwen3.6-max-preview`	94	21.6	12.7	40	41.2
`gpt-5.6-sol`	133	20.9	8.7	55	58.5
`claude-opus-5`	104	20.7	10.6	45	48.9
`deepseek-v4-flash-0731`	93	19.8	10.9	40	44.9
`qwen3.6-plus`	99	19.8	10.2	43	48.3
`glm-5.2`	86	18.7	9.6	41	48.8
`gpt-5.6-terra`	100	18.7	9.8	40	47.6
`gpt-5.6-luna`	112	18.4	8.4	45	54.2
`claude-sonnet-5`	97	18.0	9.8	37	45.7
`gemini-3.1-flash-lite`	95	17.1	8.7	38	49.4
pooled	—	—	—	471	**48.3**

Table 12: What the undecidability convention is worth. "Proven-EQ only" requires a demonstrated equivalence rather than the absence of a demonstrated difference; the final columns are the tasks that separate the two.

48% of all USABLE@3 credit — 471 of 975 task-credits — rests on a verdict the engine could not reach rather than on demonstrated equivalence, and the share tracks UNDEC count closely across models ($r = +0.82$) even though the composite itself does not. Removing the credit reorders the field. `gpt-5.6-sol` has the most undecidable comparisons of any model at 133; it has the third-highest USABLE@3 and one of the three lowest proven-equivalence rates, and it falls seven places.

We are not proposing the stricter rule — it would charge a model for a theorem, which is the objection to DFA-EQ — and we are not claiming any model games this deliberately. The point is narrower and it is a third independent reason to distrust the composite: nearly half of what USABLE certifies, it certifies by default.

5.3 Paired inference

Our first analysis compared systems with independent binomial confidence intervals. That is the wrong estimator for this design. All eleven models answer the same items, so task difficulty is a shared random effect. Treating the systems as independent samples charges every one of them separately for the same variation.

A paired bootstrap fixes this, and the fix is not ours: the procedure is Berg-Kirkpatrick et al. [2012], following Koehn [2004], and Dror et al. [2018] recommend it for exactly this design. Resample tasks with replacement [Efron and Tibshirani, 1994], score every model on each resample, and form the per-pair differences. Task difficulty cancels, and only disagreement between the two systems contributes. With $B = 10{,}000$ resamples and a fixed seed:

Procedure	pairs resolved at 95% (of 55)
Independent binomial intervals	1
Paired bootstrap, USABLE@3	9
Paired bootstrap, PASS@3	15
Paired bootstrap, VULNERABLE@3	15

Table 13: Resolvable pairwise comparisons under independent versus paired inference on identical data.

The correction is not marginal. It changes the number of defensible comparisons ninefold. We flag it because independent intervals look like the default in adjacent evaluation work, and the design they are applied to, all systems answering all items, is close to universal.

Even corrected, the middle of the table does not separate. For example `qwen3.6-max-preview` versus `gpt-5.6-sol` yields +0.7% with a 95% interval of $[-2.9\%, +4.3\%]$.

5.4 Discriminative power of the corpus

There is a structural reason behind this. On USABLE@3, **62% of tasks produce the same outcome for all eleven models**. On PASS@3 it is 56%, on VULNERABLE@3, 77%. Only 167 of 450 tasks carry any discriminative signal at all.

For calibration: resolving a 3-percentage-point difference at $p \approx 0.2$ with independent intervals requires roughly 2,800 items—more than the full 762-task corpus contains. Paired inference reduces this requirement substantially but cannot manufacture signal from items on which every system agrees. Benchmark construction for system comparison should therefore optimise for discriminative yield, not corpus size.

6 Threats to Validity

Contamination. The corpus predates every model we evaluated and is public, as are the forum posts it was built from. Some of what we measured may be memorisation. Sainz et al. [2023] argue that this has to be measured per benchmark rather than assumed negligible, and we cannot measure it: we have no held-out private task set, so we cannot put a bound on it. This is the largest unaddressed threat in the paper.

Corpus-local effect sizes. Section 4.6 replicates PASS and VULNERABLE on a second corpus and finds the vulnerability rate among correct patterns differs by a factor of 2.2. Every effect size

in this paper should therefore be read as a property of Re(gEx|DoS)Eval rather than of natural-language-to-regex generation. DFA-EQ and EXACT remain single-corpus: replicating those needs a second corpus with a reference in standard syntax, and neither KB13, NL-RX nor StructuredRegex supplies one. A further hazard we avoided rather than solved: KB13 and NL-RX are written in a DSL where `&` is intersection and `~` is negation, both of which Python's `re` accepts silently as literals. We excluded the affected patterns rather than mistranslate them: NL-RX-Synth goes from 10,000 rows to 9,648 distinct targets to the 5,840 screened, and KB13 from 824 rows to 732 distinct to 532. (An earlier draft gave the exclusion counts over raw rows and the pool sizes over distinct ones, which did not reconcile.)

Single prompt. Every result here is conditional on one unoptimised prompt. Section 5.1 shows how often the task descriptions leave the anchoring convention unstated, so sensitivity to phrasing is plausibly large. We did not measure it.

Sampling control. We do not set temperature (Section 3.5), so it varies by provider default. We checked that outputs still vary across samples. We did not equalise the sampling.

Estimator degeneracy. At $n = k = 3$ the PASS@k estimator reduces to any-of-3.

Coverage asymmetry. Nine models cover all 450 tasks. `kimi-k3` covers 447, having exhausted our budget mid-run, and `claude-opus-5` covers 444, losing five tasks to content-filter refusals and one to replies that were nothing but a code fence (Section A). Tasks that came back with fewer than k samples are excluded from that model's @k estimate rather than scored on what arrived (Section 3.4), so the denominators in Table 1 are 432–450.

Screening, not proof. VULNERABLE is a lower bound (Section 3.4).

Single corpus. We do not test whether the observed banding survives corpus change.

Self-adjudication. See Section 5.1.

7 Recommendations for Evaluation Design

1. **Partition metrics by reference dependence, and decompose any composite that spans the partition.** Metrics evaluated by execution against inputs (PASS, VULNERABLE) are unaffected by reference-set defects. Metrics evaluated against a human answer key inherit its error rate. A composite conjoining both can report a large gap that is almost entirely attributable to the unreliable term—in our case 87% of it (Section 4.2)—while appearing to be a statement about the reliable ones.

2. **Sample and adjudicate disagreements.** It costs hours. In our case it changed the conclusion.

3. **Use paired inference when all systems answer all items.** Table 13.

4. **Report discriminative yield**, not only corpus size (Section 5.4).

5. **Pin the serving provider and verify post hoc.** Record what served each request, not only what was requested (Section 3.3).

6. **Instrument the scorer with controls** that must pass and must fail (Section 3.4).

7. **Release raw generations.** Anyone can then overturn a scoring decision without buying the inference again.

8. **Treat an effect size as corpus-local until replicated.** Our reference-independent gap is 7.4% on one corpus and 16.5% on another under an identical screen (Section 4.6). Reference-independence buys freedom from answer-key error. It does not buy generality.

9. **Check that a partial run is not a biased run.** When a model loses tasks to infrastructure, score the remaining models on the tasks it lost. Ours were 8.1 points harder (Section B), which would have flattered the affected model rather than penalised it. Check separately that the *estimator* is defined on the partial data: ours was not, and credited every short task to whichever model lost it (Section B).

10. **Generate every table from committed data, and let the build fail.** The reviewer of an earlier draft of this paper found nine arithmetic inconsistencies. Every one of them was in a table we maintained by hand or in a sentence quoting one, and none was in a table generated from the scores. The fix is mechanical rather than editorial: emit all of them, keep the numbers the prose states in macros generated alongside, and have continuous integration rebuild and diff.

11. **Calibrate a screen per population before comparing populations.** A detector's false-negative rate is a property of the patterns it is pointed at. Reporting it as a single caveat is adequate for one population and lets a differential miss rate masquerade as a finding across several (Section 4.4.2).

8 Conclusion

We set out to measure a gap between correct and shippable regular expressions. We built a composite metric, it showed a large gap, and decomposition showed that 87% of that gap came from the one term we had already shown to be unreliable. What survives is much smaller. 7.4% of functionally correct patterns carry a ReDoS vulnerability, against roughly 50% of functionally correct backends in comparable work. The correctness-to-security penalty depends on the domain, and in ours it is small.

Two findings survive that correction intact, both resting only on criteria that never consult a human answer key. Models are safer than the human-written reference patterns against which they are scored — on eight of eleven under a paired test, not on all eleven, which is what we would have claimed from the point estimates — and indistinguishable from the regular expressions in shipped packages once both are restricted the same way. Across six populations, ReDoS prevalence tracks whether a pattern is ever executed rather than whether a human or a model wrote it, which locates the anomaly in the answer key rather than in generation or in human practice. And the entire eleven-model field lies within seven percentage points across a 98× range in inference cost.

A third check went the same way. The 7.4% that survived the first two corrections is 16.5% on a second corpus under the same screen and the same models (Section 4.6), so the number we were left defending is corpus-local too.

All three corrections ran the same way. A composite agreed with the literature and turned out to be dominated by its weakest term; a human baseline on one corpus supported a strong claim that a second population refuted; a reference-independent rate proved to be a fact about the corpus. In each case the check was cheap and we nearly skipped it. The recurring lesson is not about regular expressions. It is that a number which agrees with what you expected is the one most worth spending an afternoon attacking.

The methodological result is the one we did not plan on. A metric that compares against a human answer key inherits that key's error rate, and a composite built on such a term can produce a finding that is large, plausible and mostly spurious. We only found this by reading our own output after collecting it. Fourteen cases were enough to change what we were willing to publish off a $10.95 run. We would read them earlier next time.

Acknowledgements

Scoring is performed by `regexbench` [Plicara Labs, 2026a], prior work by the present authors, used here as a pinned dependency and not contributed by this paper. The corpus is Re(gEx|DoS)Eval [Siddiq et al., 2024a], used under its original terms and not redistributed.

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A Failure taxonomy

Of 14,850 requests, 54 (0.36%) failed and are reported rather than excluded.

Model	Cause	Count
`claude-opus-5`	content-filter refusal	29
`claude-opus-5`	code fence, no pattern inside	7
`kimi-k3`	account spending limit reached mid-run	11
`kimi-k3`	response without resolved provider	3
`kimi-k3`	code fence, no pattern inside	2
`gpt-5.6-sol`	response without resolved provider	1
`gpt-5.6-luna`	code fence, no pattern inside	1
	total	*54*

Table 14: Request failures by cause. Responses lacking a resolved provider are discarded rather than scored, as provenance is a precondition for reproducibility. The code-fence rows are responses the API returned as successful, from which extraction yields the empty string: nine emitted a fence with nothing inside it beyond an optional language tag, and one wrote a pattern followed by an unmatched closing fence, which leaves the last fenced block empty. All ten are short replies of 2–18 completion tokens; none is a truncation against the 400-token cap. An earlier draft of this table omitted them and accounted for only 44 of the 54.

Empty completions are classified by reported `finish_reason` into refusal, token-budget exhaustion, and unexplained, rather than being scored as empty patterns—which would be indistinguishable from a maximally poor answer.

B Operational hazards

We document configuration hazards encountered during this study, each of which produced plausible-looking but invalid data. We believe this list has independent value.

1. **Parameter combinations that are silently unroutable.** Setting `temperature` together with `require_parameters=true` yields a hard routing failure on 6 of 11 models. Our first pilot failed all 396 calls.

2. **Systems that cannot satisfy the experimental condition.** Two Gemini variants return *"Reasoning is mandatory for this endpoint"* and cannot be evaluated with reasoning disabled. Undetected, this would have placed one reasoning-enabled system in an otherwise reasoning-disabled comparison.

3. **Empty completions from budget exhaustion.** At a 200-token cap, reasoning models consumed the entire budget on hidden reasoning and returned no content.

4. **Resumption across configuration changes.** Our collection is resumable by (model, task, sample). An aborted run at one token cap left truncated rows that a resumed run at a different cap treated as complete, interleaving two configurations indistinguishably. Rows now carry a configuration fingerprint and resumption refuses on mismatch.

5. **Sample collapse in scoring.** An early scorer retained only the final sample per task, which would have rendered all @3 statistics disguised @1 statistics with no visible symptom.

6. **An estimator guard that is sound only on its own domain.** The PASS@k estimator is usually implemented with the shortcut "if $n - c < k$, return 1": if fewer than k samples failed, every choice of k must contain a success. That is correct for $n \geq k$, which is the only case the estimator is defined on, and it is silently wrong below it. A task that returned one sample satisfies $n - c \leq n < k$ whatever happened, so it scores a full 1.0 on every metric — pass_at_k$(1, 0, 3) = 1$ — and a task nobody answered correctly is counted as answered correctly.

   We found this late, from a discrepancy of 1.3 points between two of our own tables, and it had been inflating exactly the two models that lost the most samples, which were the two at the top of the table. Correcting it moves `claude-opus-5` from second to fourth on USABLE@3 and makes `kimi-k3` rather than `claude-opus-5` the highest PASS@3. No claim in this paper turns on it — we publish no ranking, and Section 5.4 already says the band does not separate — but Table 1 is ordered by the affected column, and a reader comparing two tables would have had no way to tell which was right.

   The general form is worth naming: a fast path justified by a precondition, in code that is also called when the precondition does not hold. The failure is silent, it is biased toward the systems with the most missing data, and nothing in the output looks wrong.

A content filter can silently bias a subset. Collecting StructuredRegex, Anthropic's content filter rejected 182 of 622 descriptions for `claude-opus-5`, all of them benign ("A string that starts with one or more digits and optionally ends with 'NU' or 'DG'."). A probe suggested the filter was stochastic: 8 of 20 rejected prompts succeeded when resent unchanged. Five retry rounds recovered 98, with the per-round yield collapsing (51, 31, 9, 2, 5), which shows a stochastic component over a hard core the filter rejects every time. That leaves 84 prompts the filter rejected on every attempt. A further 25 responses came back successful but yielded no pattern, and are not the filter's doing: 24 are a bare code fence of one or two tokens, the same failure mode as Section A's, and one is a genuine truncation, having spent the whole 400-token budget inside a `<thinking>` block. Together $84 + 25 = 109$, and final coverage was $622 - 109 = 513$.

The retry effort was not uniform across models, because nothing else needed it: only `claude-opus-5` was blocked, so only `claude-opus-5` received five extra rounds. The recovered 98 are therefore conditioned on non-refusal in a way the other ten models' answers are not. We do not think this biases regex quality — the filter keys on the description, which the model has not yet answered — but it is differential effort and the common-subset comparison below is the reason it does not matter for the reported numbers.

The recovery is not the interesting part. Scoring the other ten models on the 109 blocked tasks against the 513 kept shows the blocked tasks are 8.1 points harder on average (sd 3.3, negative for all ten). The surviving subset therefore *flattered* the affected model, at the same time as counting the blocked rows as failures *penalised* it. Either error alone is visible; together they produce a plausible number that means nothing. We report the common 513-task subset for all eleven models (Table 8) and release `runner/filter_bias_check.py`, which needs no API access.

C Reproduction

```
git clone https://github.com/plicara/regexeval-2026
cd regexeval-2026
make setup # pinned scorer + corpus download
```

```
make score RUN=sweep
```

Scoring reads only the committed generations. No API access, no expenditure. We verified reproduction from a clean checkout with fresh dependency installation and corpus download, obtaining bit-identical metrics. A reduced form of this check executes in continuous integration on every commit.